

```
print("Computer Programming 2026\n")  
print("Department of Geography\n")  
print("2 April 2026\n")  
  
print("Week 6")
```

List

- Group of consecutive cells
 - Representation of variable groups
 - With the same name (but not necessarily the same type)
- To refer to an ***element***, we must specify:
 - The name of the list
 - The ***position*** of the element (index)
- Syntax: *list_name*[*element_position*]
 - The first element has index **0**
 - The n-th element of the list **v** is **v[n-1]**

list name

v[0]	-45
v[1]	6
v[2]	0
v[3]	72
v[4]	1543
v[5]	-89
v[6]	0
v[7]	62
v[8]	-3
v[9]	1
v[10]	6453
v[11]	78

element position

Exercise

```
data = []
DIM = 100

for i in range(DIM):
    data.append(int(input(f"Insert the number (choice {i+1}): ")))

print("\n")
for i in range(DIM):
    print(f"The number (choice {i+1}) is: {data[i]}")

print("\n")
for i in range(DIM, 0, -1):
    print(f"The number (choice {i}) is: {data[i-1]}")
```

A FEW NOTES...

A few notes...

- Create a list: `mylist = ["apple", "banana", "cherry"]`

- Lists allow duplicate values:

```
mylist = ["apple", "banana", "cherry", "apple", "cherry"]
```

- Print the number of items in the list: `print(len(mylist))`

- You can also use the `list()` constructor to create a list:

```
mylist = list(("apple", "banana", "cherry"))
```

A few notes...

- List of strings: `list1 = ["apple", "banana", "cherry"]`
- List of integers: `list2 = [1, 5, 7, 9, 3]`
- List of boolean values: `list3 = [True, False, False]`
- List with strings, integers, and boolean values:
`list4 = ["abc", 34, True, 40, "male"]`

A few notes...

- Print the second item of the list:

```
mylist = ["apple", "banana", "cherry"]  
print(mylist[1])
```

- Print the last item of the list:

```
mylist = ["apple", "banana", "cherry"]  
print(mylist[-1])
```

- Print the third, fourth and fifth item of the list:

```
mylist=["apple","banana","cherry","orange","kiwi","melon","mango"]  
print(mylist[2:5])
```

A few notes...

- Print the first four items of the list:

```
mylist=["apple","banana","cherry","orange","kiwi","melon","mango"]  
print(mylist[:4])
```

- Print the last five items of the list:

```
mylist=["apple","banana","cherry","orange","kiwi","melon","mango"]  
print(mylist[2:])
```

- Print the fourth, fifth and sixth item in the list:

```
mylist=["apple","banana","cherry","orange","kiwi","melon","mango"]  
print(mylist[-4:-1])
```


A few notes...

- Check if “apple” is present in the list:

```
mylist = ["apple", "banana", "cherry"]  
if "apple" in mylist:  
    print("Yes, 'apple' is in the fruits list")
```

- Change the second item of the list:

```
mylist = ["apple", "banana", "cherry"]  
mylist[1] = "strawberry"
```

- Change the values “banana” and “cherry” with the values “strawberry” and “watermelon”:

```
mylist=["apple", "banana", "cherry", "orange", "kiwi", "melon", "mango"]  
mylist[1:3] = ["strawberry", "watermelon"]
```

A few notes...

- Change the second value by replacing it with two new values:

```
mylist = ["apple", "banana", "cherry"]  
mylist[1:2] = ["strawberry", "watermelon"]
```

- Change the second and third value by replacing it with one value:

```
mylist = ["apple", "banana", "cherry"]  
mylist[1:] = ["watermelon"]
```

- Insert "watermelon" as the third item:

```
mylist = ["apple", "banana", "cherry"]  
mylist.insert(2, "watermelon")
```

A few notes...

- Insert an item at the end:

```
mylist = ["apple", "banana", "cherry"]  
mylist.append("orange")
```

- Insert an item in the second position:

```
mylist = ["apple", "banana", "cherry"]  
mylist.insert(1, "orange")
```

- Add the elements of *tropical* to *mylist*:

```
mylist = ["apple", "banana", "cherry"]  
tropical = ["mango", "pineapple", "papaya"]  
mylist.extend(tropical)  
  
mylistnew = mylist + tropical
```

A few notes...

- Remove “banana”:

```
mylist = ["apple", "banana", "cherry"]  
mylist.remove("banana")
```

- Remove the first occurrence of “banana”:

```
mylist = ["apple", "banana", "cherry", "banana", "kiwi"]  
mylist.remove("banana")
```

- Remove the second item:

```
mylist = ["apple", "banana", "cherry"]  
mylist.pop(1)
```

A few notes...

- Remove the last item:

```
mylist = ["apple", "banana", "cherry"]  
mylist.pop()
```

- Remove the first item:

```
mylist = ["apple", "banana", "cherry"]  
del mylist[0]
```

- Delete the entire list:

```
mylist = ["apple", "banana", "cherry"]  
del mylist
```

- Clear the list content:

```
mylist = ["apple", "banana", "cherry"]  
mylist.clear()
```

A few notes...

- Sort the list alphabetically:

```
mylist = ["orange", "mango", "kiwi", "pineapple", "banana"]  
mylist.sort()
```

- Sort the list descending:

```
mylist = ["orange", "mango", "kiwi", "pineapple", "banana"]  
mylist.sort(reverse = True)
```

- Sort the list numerically:

```
mylist = [100, 50, 65, 82, 23]  
mylist.sort()
```

- Sort the list descending :

```
mylist = [100, 50, 65, 82, 23]  
mylist.sort(reverse = True)
```

- Reverse the order:

```
mylist.reverse()
```

A few notes...

- Copy a list:

```
mylist = [100, 50, 65, 82, 23]
```

```
list2 = mylist
```

A few notes...

- Copy a list:

```
mylist = [100, 50, 65, 82, 23]
```

~~list2 = mylist~~ **WRONG**

`newlist = mylist.copy()` **CORRECT**

`newlist = list(mylist)` **CORRECT**

A few notes...

- Increase all values in the list by 1:

```
mylist = [100, 50, 65, 82, 23]
```

```
mylist = mylist + 1
```

A few notes...

- Increase all values in the list by 1:

```
mylist = [100, 50, 65, 82, 23]
```

~~`mylist = mylist + 1`~~ **WRONG**

```
for i in range(len(mylist)):  
    mylist[i] = mylist[i] + 1
```

CORRECT

```
for i in range(len(mylist)):  
    mylist[i] += 1
```

CORRECT

Exercise

Problem

- There are 10 judges
 - 1 judge = 1 vote
- We want to memorize the 10 votes and print them out
- Each vote must fall within the interval 0-10
- Given a dive, calculate:
 - the average of the votes
 - the highest vote and the lowest vote

Exercise

Problem

- There are 10 judges
 - 1 judge = 1 vote
- **We want to memorize the 10 votes and print them out**
- Each vote must fall within the interval 0-10
- Given a dive, calculate:
 - the average of the votes
 - the highest vote and the lowest vote

Exercise (old version)

```
judge = 1
high = 0
low = 10
tot = 0

while judge <= 10:
    vote = int(input(f"Insert the vote of judge {judge}: "))
    if vote > high:
        high = vote
    if vote < low:
        low = vote
    tot = tot + vote
    judge += 1

print("\nThe highest vote is", high)
print("\nThe lowest vote is", low)
print("\nThe average is", tot / 10)
```

Exercise

Variables:

```
high = 0
low = 10
tot = 0
vote = []
```

- Problem 1: save the data into the list (vote=[])
- Problem 2: define a loop that allows going through the elements of the list

Exercise

- Problem 1: save the data into the list (vote=[])

```
vote.append(input("...: "))
```

- Problem 2: define a loop that allows going through the elements of the list

```
for i in range(10):
```


Exercise

- Problem 1 + Problem 2

```
for i in range(10):  
    vote.append(input("...: "))
```

Exercise

- Data entry

```
for judge in range(10):  
    vote.append(int(input(f"Insert the vote of judge {judge+1}: ")))  
    if vote[judge] > high:  
        high = vote[judge]  
    if vote[judge] < low:  
        low = vote[judge]  
    tot = tot + vote[judge]
```

Exercise

- Print output

```
for judge in range(10):  
    print(f"The vote of judge {judge+1} is {vote[judge]}")  
  
print("\nThe highest vote is", high)  
print("\nThe lowest vote is", low)  
print("\nThe average is", tot / 10)
```

- All together

```
high = 0
low = 10
tot = 0
vote = []

for judge in range(10):
    vote.append(int(input(f"Insert the vote of judge {judge+1}: ")))
    if vote[judge] > high:
        high = vote[judge]
    if vote[judge] < low:
        low = vote[judge]
    tot = tot + vote[judge]

print("\n")

for judge in range(10):
    print(f"The vote of judge {judge+1} is {vote[judge]}")

print("\nThe highest vote is", high)
print("\nThe lowest vote is", low)
print("\nThe average is", tot / 10)
```

Exercise

```
vote = []

for judge in range(10):
    vote.append(int(input(f"Insert the vote of judge {judge+1}: ")))

print("\n")

for judge in range(10):
    print(f"The vote of judge {judge+1} is {vote[judge]}")

print("\nThe highest vote is", max(vote))
print("\nThe lowest vote is", min(vote))
print("\nThe average is", sum(vote) / 10)
```


**...AND IF WE WANT TO USE
LISTS WITH FUNCTIONS...?**

WE ALREADY KNOW!

Parameter transfer (list)

- At calling time, the current parameters are linked to the formal parameters
 - the current parameter and the formal parameter refer to the same **memory cell**
- The subprogram, during execution, works in its environment on the formal parameters (and consequently also on the current parameters)
 - every modification on the formal parameter is a modification of the corresponding current parameter
- The effects of the subprogram arises in the caller, with modifications in its environment

**WHY DOESN'T THE FUNCTION
MAKE AN INTERNAL COPY OF THE
LIST, INSTEAD OF WORKING ON
THE ORIGINAL LIST?**

**DOES IT MAKE SENSE
TO COPY A LIST?**

NO!

**IF A LIST HAS 100000000
ELEMENTS...?**

Exercise

- After saving a sequence of N integers inserted by the user, print out on screen the sum of these N numbers

(DON'T use default functions! Create a function by yourself!)

Exercise

- **After saving a sequence of N integers inserted by the user, print out on screen the sum of these N numbers**

```
inputlist = []  
  
a = int(input("How many numbers do you want to insert? "))  
  
for i in range(a):  
    inputlist.append(int(input(f"Insert the number (choice {i+1}): ")))
```

Exercise

- After saving a sequence of N integers inserted by the user,
print out on screen the sum of these N numbers
(create a function)

```
result = sumFun(inputlist)
```

```
def sumFun(mylist):  
  
    result = 0  
  
    for i in range(len(mylist)):  
        result = result + mylist[i]  
  
    return result
```

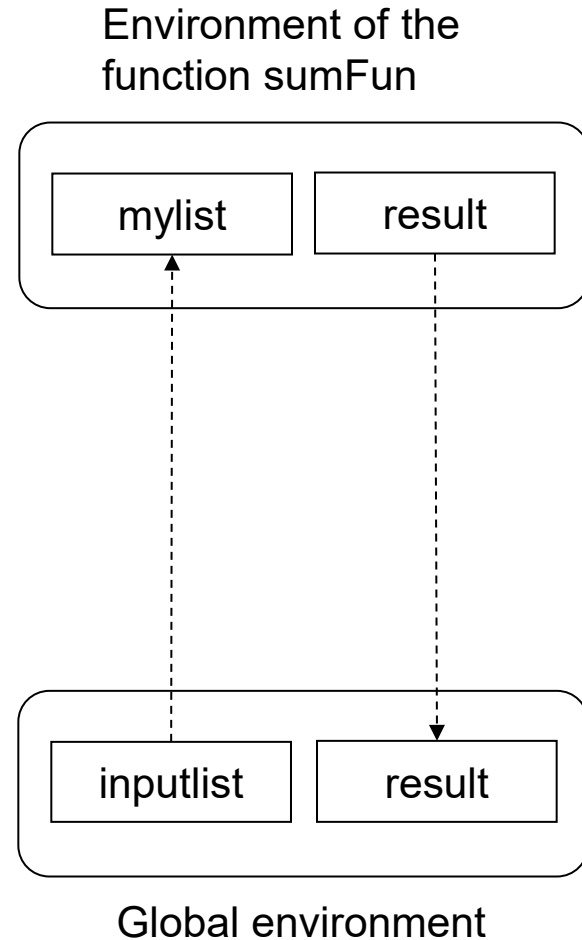
Exercise

```
def sumFun(mylist):  
    result = 0  
    for i in range(len(mylist)):  
        result = result + mylist[i]  
    return result  
  
inputlist = []  
a = int(input("How many numbers do you want to insert? "))  
  
for i in range(a):  
    inputlist.append(int(input(f"Insert the number (choice {i+1}): ")))  
  
result = sumFun(inputlist)  
print(f"\nThe sum of the {a} numbers is: {result}")
```

Exercise

```
def sumFun(mylist):  
    result = 0  
  
    for i in range(len(mylist)):  
        result = result + mylist[i]  
  
    return result
```

```
result = sumFun(inputlist)
```



ARE THEY THE SAME THING?

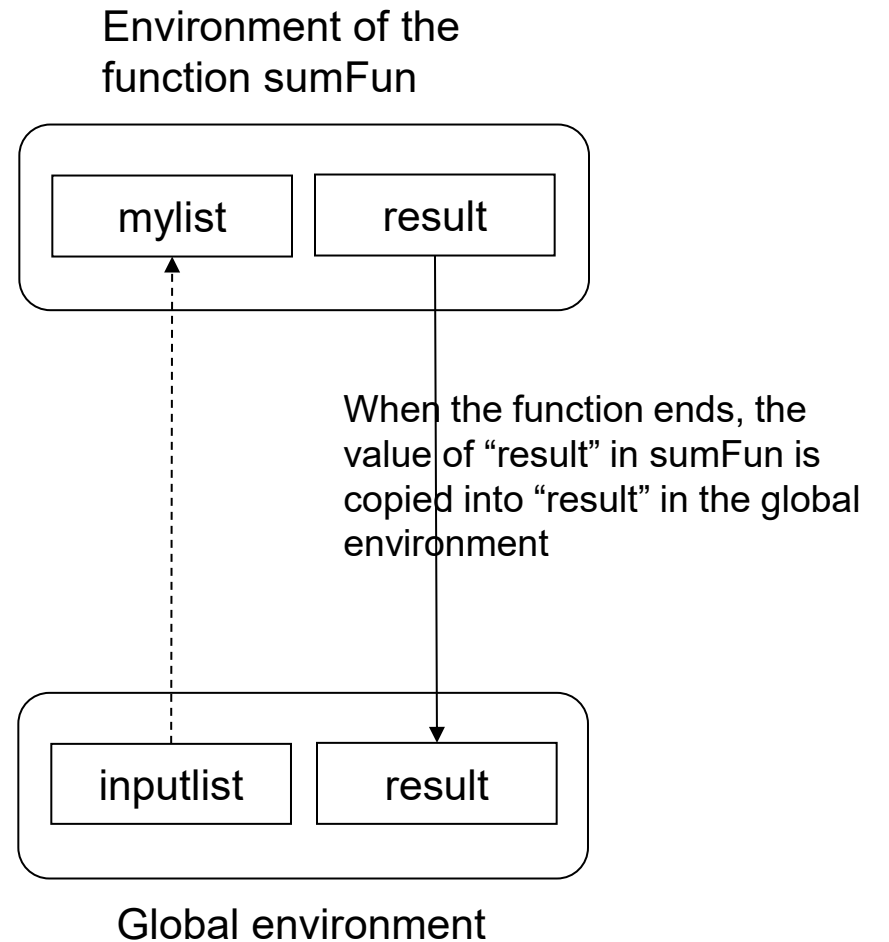
VISIBILITY *EFFECT*!



Exercise

```
def sumFun(mylist):  
    result = 0  
  
    for i in range(len(mylist)):  
        result = result + mylist[i]  
  
    return result
```

```
result = sumFun(inputlist)
```

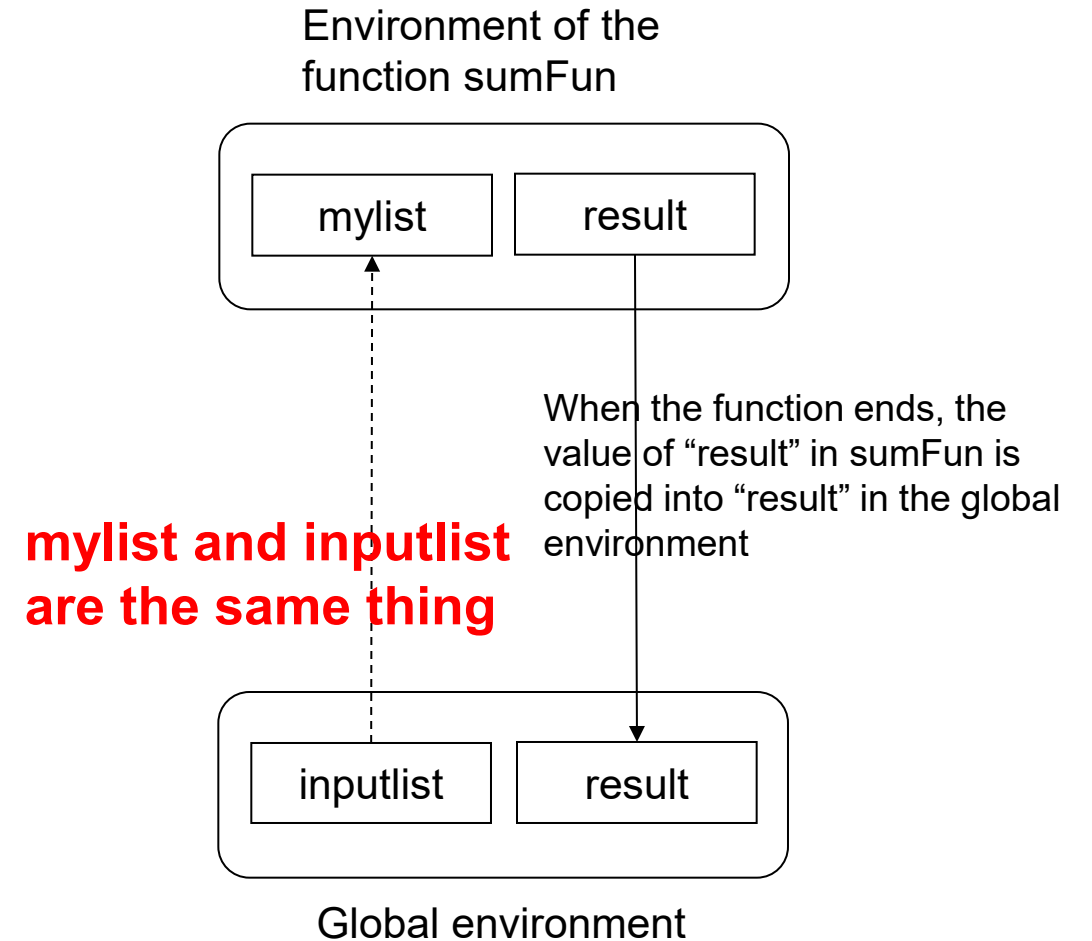


ARE THEY THE SAME THING? **NO!**

Exercise

```
def sumFun(mylist):  
    result = 0  
  
    for i in range(len(mylist)):  
        result = result + mylist[i]  
  
    return result
```

```
result = sumFun(inputlist)
```



Exercise

- Create a function that, given an input list of N integers, calculates the sum of those numbers and appends the result at the end of the list

```
def sumFun_v2(mylist):  
  
    mylist.append(0)  
  
    for i in range(len(mylist)-1):  
        mylist[-1] = mylist[-1] + mylist[i]  
  
    return
```

Exercise

```
def sumFun_v2(mylist):  
    mylist.append(0)  
    for i in range(len(mylist)-1):  
        mylist[-1] = mylist[-1] + mylist[i]  
    return  
  
inputlist = []  
a = int(input("How many numbers do you want to insert? "))  
  
for i in range(a):  
    inputlist.append(int(input(f"Insert the number (choice {i+1}): ")))  
  
sumFun_v2(inputlist)  
print(f"\nThe sum of the {a} numbers is: {inputlist[-1]}")
```

Exercise (recursion)

- Design a recursive procedure to calculate the maximum of a list of integers

Idea:

$$\text{maximum}(\text{list}[0:N]) = \text{maximum}(\text{list}[0], \text{maximum}(\text{list}[1:N]))$$

Exercise (recursion)

```
def maximum(mylist):  
    if len(mylist) == 2: # basic case  
        if mylist[0] > mylist[1]:  
            maxi = mylist[0]  
        else:  
            maxi = mylist[1]  
    else: # solving reduced problem  
        maxi = maximum(mylist[1:])  
        if mylist[0] > maxi:  
            maxi = mylist[0]  
    return maxi
```

See you
next week!